CONCEPTUAL PHYSICS AND CHEMISTRY IN DAILY LIFE

ACADEMIC

Grade 12

Curriculum Guide

CONCEPTUAL PHYSICS AND CHEMISTRY IN DAILY LIFE
Prerequisite: General Science (Core Subject)

Course Description:

This course introduces learners to key ideas in physics and chemistry and how these concepts relate to common everyday experiences. Through a conceptual approach, the learners explore how principles of motion, aerodynamics, light, waves, chemical reactions, gases, and organic substances influence technologies, household products, environmental issues, and modern industries.

By connecting scientific concepts to familiar situations such as transportation, food preparation, medicine, communication technologies, and environmental challenges, the learners develop the ability to interpret everyday phenomena using scientific reasoning and appreciate the role of physics and chemistry in improving quality of life.

Elective: Academic
Time Allotment: 80 hours for one term

CONTENT	**CONTENT STANDARDS** *The learners demonstrate understanding that:*	**LEARNING COMPETENCIES** *The learners:*
Conceptual Physics in Daily Life		
1. Physicists and Careers in Physics	1. scientific discoveries in physics and chemistry are shaped by the contributions of scientists and lead to innovations that improve everyday life and create diverse career opportunities in science and technology.	1. identify Filipino and international physicists and explain how their discoveries contribute to technologies used in everyday life;
		2. identify and explore various career paths and opportunities related to Physics (e.g. Medical Physics, Biophysics, Nuclear Physics, Meteorology, Space Science, Data Science, Materials Science, Engineering, and Energy Studies);

CONTENT	CONTENT STANDARDS *The learners demonstrate understanding that:*	LEARNING COMPETENCIES *The learners:*
2. Circular and Rotational Motion in Daily Life	2. principles of motion, including circular and rotational motion, explain the movement of objects and systems and guide the design and safe use of machines, transportation systems, and other technologies in everyday life.	3. describe the connected concepts of circular and rotational motion in everyday life, such as satellite motion, amusement park rides, and safety practices in driving cars and handling industrial machines;
3. Aerodynamics and Air Pressure in Design	3. aerodynamic principles involving air pressure and airspeed influence the behavior of objects and are applied in the design of vehicles, structures, and technologies to improve efficiency and performance; and	4. describe how aerodynamic objects such as umbrellas, roofs and airplane wings behave in situations with unequal air speeds and air pressures;
		5. gather information through secondary sources how aerodynamic vehicles and structures are designed based on the effects of airspeed and air pressure;
4. Light, Color, and Wave Interference	4. light and wave phenomena, including color formation and wave interference, explain observable patterns in nature and are applied in technologies used in communication, engineering, entertainment, and industry.	6. explore the applications of color mixing with pigments and light in industries such as art, entertainment, photography, engineering, and food production;
		7. investigate the phenomenon of wave interference in water ripples, vibrating guitar strings, sound amplifiers; and overlapping radio signals; and
		8. use secondary sources to discuss applications of wave interference in technologies such as noise canceling devices, holography, and anti-reflection coatings.

CONTENT	CONTENT STANDARDS *The learners demonstrate understanding that:*	LEARNING COMPETENCIES *The learners:*
Performance Standards *The learners* can apply key concepts of motion, aerodynamics, light, and wave behavior to explain everyday phenomena and technological innovations. They can analyze how scientific discoveries and careers in Physics contribute to societal development and demonstrate how these principles are applied in real-life technologies and industries.		
Suggested Performance Task Design and present a physical model demonstrating how principles of circular or rotational motion or aerodynamics are applied in a real-world technology such as amusement park rides, satellite motion, vehicle safety systems, airplane wings, or roof and umbrella designs.		
Conceptual Chemistry in Daily Life		
5. Atomic and Molecular Mass in Everyday Products	5. atomic mass, formula mass, and molecular mass help determine the composition and proper use of substances in daily life such as in nutrition, medicine, and chemical products;	9. explain the significance of calculating atomic mass, formula mass, and molecular mass in everyday applications such as nutrition labeling, pharmaceutical dosage formulation, and chemical product development;
6. Mole Concept and Counting Particles	6. the mole concept is used by scientists to count very small particles and is applied in medicine, environmental monitoring, and industry;	10. explain the mole concept in counting particles of a substance and describe its applications in medicine, environmental monitoring, and industry including pharmaceuticals and pollution monitoring;
7. Rates of Chemical Reactions in Everyday Life	7. factors such as surface area and temperature affect how fast chemical reactions occur and influence the effectiveness and shelf life of many products;	11. investigate how factors affecting the rates of chemical reactions, such as surface area and temperature, influence real-life applications, including cooking, food spoilage, drug efficacy and the shelf life of consumer products;

CONTENT	**CONTENT STANDARDS** *The learners demonstrate understanding that:*	**LEARNING COMPETENCIES** *The learners:*
8. Organic Compounds in Daily Life	8. organic compounds are present in many materials at home and in school and are widely used in consumer products, industry, health, and medicine;	12. describe different organic compounds that can be found at home and in school and their role in consumer products, industry, health and medicine;
9. Environmental Impact of Organic Compounds	9. some organic compounds can affect the environment and contribute to issues such as air pollution and climate change; and	13. explain how specific organic compounds contribute to environmental issues such as air pollution and climate change; and
10. Relationships among Pressure, Volume, and Temperature	10. changes in pressure, volume, and temperature affect gases and explain many everyday applications and technologies.	14. conduct observations and interpret data to describe how changes in: a. pressure affects the volume of a gas, b. temperature affects the volume of a gas, and c. temperature affects the pressure of gas.
Performance Standards *The learners* can explain how basic chemistry concepts such as particle measurement, reaction rates, organic compounds, and gas relationships are applied in everyday life. They can describe simple examples of these concepts in household products, environmental issues, and common technologies.		
Suggested Performance Task Develop a “Chemistry in Everyday Life Kit” that allows users to explore how basic chemistry concepts are applied in common household products, environmental issues, and simple technologies.		